

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION

WINTER SEMESTER, 2024–2025

DURATION: 2 HOURS

FULL MARKS: 120

SWE 4101: Introduction to Software Engineering

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) You are working on a laptop and enter a mathematical expression into the calculator application using the keyboard. After typing the expression, you press the *Enter* key to obtain the result. With the help of a neatly labeled diagram of the basic computer organization, explain how the input provided through the keyboard is processed by different components of the computer until the result is displayed on the screen. 10
(CO2)
(PO1)
- b) Increasing cache size can store more data, but may increase latency. Reducing cache latency improves access speed but limits storage. If you were designing a high-performance CPU, which would you prioritize and why? 5
(CO2)
(PO1)
- c) A software company is upgrading its computer systems to support modern applications such as video editing, virtual machines, and data analysis. Currently, some of their systems are running on older 32-bit processors, while others use 64-bit processors. The company plans to install a modern operating system and increase RAM capacity to improve performance. Based on this scenario, answer the following questions: 3 + 4 + 4 + 4
(CO2)
(PO1)
- i. What is meant by 32-bit and 64-bit processors in terms of CPU data handling?
 - ii. Compare the memory addressing capabilities of 32-bit and 64-bit processors. Why is this an important factor for the company's upgrade decision?
 - iii. Discuss the differences between data bus and address bus architecture in 32-bit and 64-bit systems and explain how these affect system performance.
 - iv. The company notices that everyday tasks such as web browsing do not show a significant performance improvement after moving to 64-bit systems, while video editing performance improves greatly. Explain the reasoning behind this.
2. a) Students in the hostels of the Institute of Unconventional Technology (IUT) often complain about slow internet speed, particularly during night hours. To address this issue, the IT department plans to design an Internet Usage Monitoring System (IUMS) 5 + 5
(CO1)
(PO1)
- i. What data should be collected about hostel internet usage to analyze the problem effectively?
 - ii. Mention five useful information that can be generated from your collected data to improve network performance and fair usage among students?
- b) A student at the Institute of Unconventional Technology (IUT) is working on an assignment using a laptop. At the same time, the student has multiple browser tabs open, a video editing software running in the background, and a music player playing audio. After some time, the system becomes very slow, stops responding to mouse clicks, and takes a long time to recover. 5 + 5
(CO1)
(PO1)
- i. Why did the computer become slow or freeze?
 - ii. Which system resources were overused and how did they affect performance?

- c) The Institute of Unconventional Technology (IUT) plans to establish an IoT and Robotics Laboratory for 80 students. The laboratory will be used for writing code, running simulations, and working with IoT devices and small robots that require real-time control of sensors and output devices (such as motors and LEDs). Identify the types of computer systems that would be required to set up this laboratory. Explain your choices with proper justification.

10
(CO1)
(PO1)

3. a) When a computer is powered on, some hardware components begin working immediately, even before the operating system loads. After the operating system starts, additional software is required so that the system can properly use hardware devices such as printers, keyboards, and display devices.

3 + 5

- Identify the two types of software referred to in the above scenario and explain their role.
- Differentiate between these two types of system software.

- b) A computer program goes through several stages from the moment it is run until it produces the final output. Identify the main stages involved in the execution of the C program given in Code Snippet 1. For each stage, explain what happens and how the program is transformed or executed.

12

```
1 #include <stdio.h>
2 #define PI 3.14    // a macro
3 int main() {
4     printf("Value of PI is %.2f", PI);
5     return 0;
6 }
```

Code Snippet 1: C program for Question 3.b

- c) Write a pseudocode that can take any octal number as input and convert it to base 6.
- d) Let the last two digits of your student ID be the value K . First, compute $K \bmod 10$, then convert this value to its binary equivalent and take the least significant bit (LSB) as D . Using this value of D , construct a complete truth table for the Boolean function given below:

5

5

$$F = (A \cdot B \cdot D) + (\neg D)$$

4. a) The Institute of Unconventional Technology (IUT) plans to set up a network connecting 50 computers across 5 labs and 2 administrative offices to share course materials, provide internet access, and run administrative applications. With proper justification, suggest the most suitable network topology for this scenario and mention one advantage and one disadvantage of using this topology in IUT's network.

5 + 2

- b) When you record your voice using a mobile phone, the microphone captures analog sound signals, but the phone stores them as digital data. Describe the steps of converting your voice from analog to digital, explaining what happens at each step.

7

- c) A student at the Institute of Unconventional Technology (IUT) attempts to recharge his IUT Wallet by logging into his account at Absolute Broke Bank Ltd. (ABBL), whose website communicates with the student's browser using the HTTP protocol. Mention the security risk posed by the use of HTTP in this scenario, propose a more secure alternative protocol, and briefly describe how this alternative protocol mitigates the identified risk.

2 +
1 + 5

- d) Explain the four main building blocks of an IoT system.

8